Quantification: 877_wt_z081c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	8
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3075.75	2935.5	2349.76	px^2
major_axis_length_px	lacuna shape	88.2344	96.7141	43.2425	px
minor_axis_length_px	lacuna shape	43.1291	45.335	17.1087	px
aspect_ratio	lacuna shape	1.9916	1.9763	0.4765	unitless
eccentricity	lacuna shape	0.8367	0.8626	0.0884	unitless
solidity	lacuna shape	0.7992	0.8371	0.1049	unitless
roots_count	headline	9.25	8	6.5629	count
ring_length_r30_px	headline	354.25	341	210.147	px
ring_length_r60_px	reported	919.625	898	439.96	px
owned_length_px	ownership-dependent	2346.36	2386.09	1771.8	px
edge_count	ownership-dependent	113.375	115.5	81.5544	count
mean_edge_length_px	ownership-dependent	19.777	20.2163	3.4096	px
perimeter_px	normalised	277.935	293.72	141.482	px
ring_area_r30_px2	normalised	9306.25	9005.5	4078.1	px^2
ring_area_r60_px2	normalised	22657	22110	8631.97	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0364156	0.036255	0.0132007	px^-1
ring_density_r60	normalised	0.0403494	0.0437166	0.0128328	px^-1
roots_per_100px_perimeter	normalised	3.2287	2.9399	1.7748	per 100 px
ring_attached_length_r30_px	network v2	312.5	295	208.281	px
ring_attached_length_r60_px	network v2	711.125	686	437.962	px
ring_length_w_r30_px	network v2	404.644	386.256	243.926	px
ring_length_w_r60_px	network v2	1058.06	1042.01	510.371	px
sholl_crossings_r10	network v2	7.875	7	5.8172	count
sholl_crossings_r20	network v2	10.75	12	5.6252	count
sholl_crossings_r30	network v2	13.625	14	7.0089	count
loop_count	loops, report only	9.625	10	6.9885	count
loop_edge_share	loops, report only	0.3074	0.3127	0.1976	share
ring_width_mean_r30_px	width	7.058	7.0847	1.0469	px

Per field
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measure	value	units
analysed_area_px2	1.0135e+06	px^2
total_skeleton_length_px	32375	px
canalicular_length_density_per_px	0.0319439	px^-1
junction_count	684	count
junction_density_per_px2	0.00067489	px^-2
lacuna_count	12	count

lacunae_per_px2	1.184e-05	px^-2
skeleton_component_count	328	count
mean_component_length_px	98.7043	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0365553	px^-1
field_density_without_flagged_per_px	0.0319439	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	502	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0319439	px^-1
field_length_density_with_unattached_per_px	0.0324392	px^-1
loop_count	99	count
loop_edge_share	0.2464	share
edge_count	1648	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4379	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	32375	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	828	47.9341	26.2785	1.8241	0.8363	0.811	1.1987	10.5652	544.853	True
2	4064	125.912	44.1728	2.8504	0.9364	0.8472	0.9481	38.4616	829.728	True
3	6500	135.386	68.4723	1.9772	0.8627	0.7101	0.7764	202.316	509.649	False
4	493	36.8679	18.6636	1.9754	0.8624	0.8203	-0.7656	201.503	648.073	False
5	4498	128.704	46.6689	2.7578	0.9319	0.9623	0.3797	397.582	25.7919	True
6	4895	134.554	48.0863	2.7982	0.934	0.8829	0.4584	399.457	314.021	False
7	1388	60.0833	43.9829	1.3661	0.6813	0.6761	-0.1762	427.803	598.192	False
8	1690	77.7941	42.0098	1.8518	0.8417	0.6483	0.0159	450.566	636.056	False
9	5044	117.23	57.6341	2.034	0.8708	0.8539	1.376	449.833	814.625	False
10	1085	58.3256	24.5501	2.3758	0.9071	0.9313	0.4295	634.654	16.1935	True
11	4181	115.634	46.6872	2.4768	0.9149	0.8959	1.0406	724.949	827.982	False
12	415	28.3253	19.4963	1.4529	0.7254	0.9061	-0.3467	864.431	580.06	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	225	625	1132.09	56	20.2158	7.3189
2	8	304	788	1030.69	45	22.9042	7.1897
3	20	720	1581	5700.64	259	22.0102	7.3764
4	7	316	816	1130.15	68	16.6198	8.5032
5	6	247	711	1027.82	44	23.3596	8.1873
6	11	431	1176	2663.89	137	19.4444	8.0356
7	3	165	543	882.749	36	24.5208	6.5979

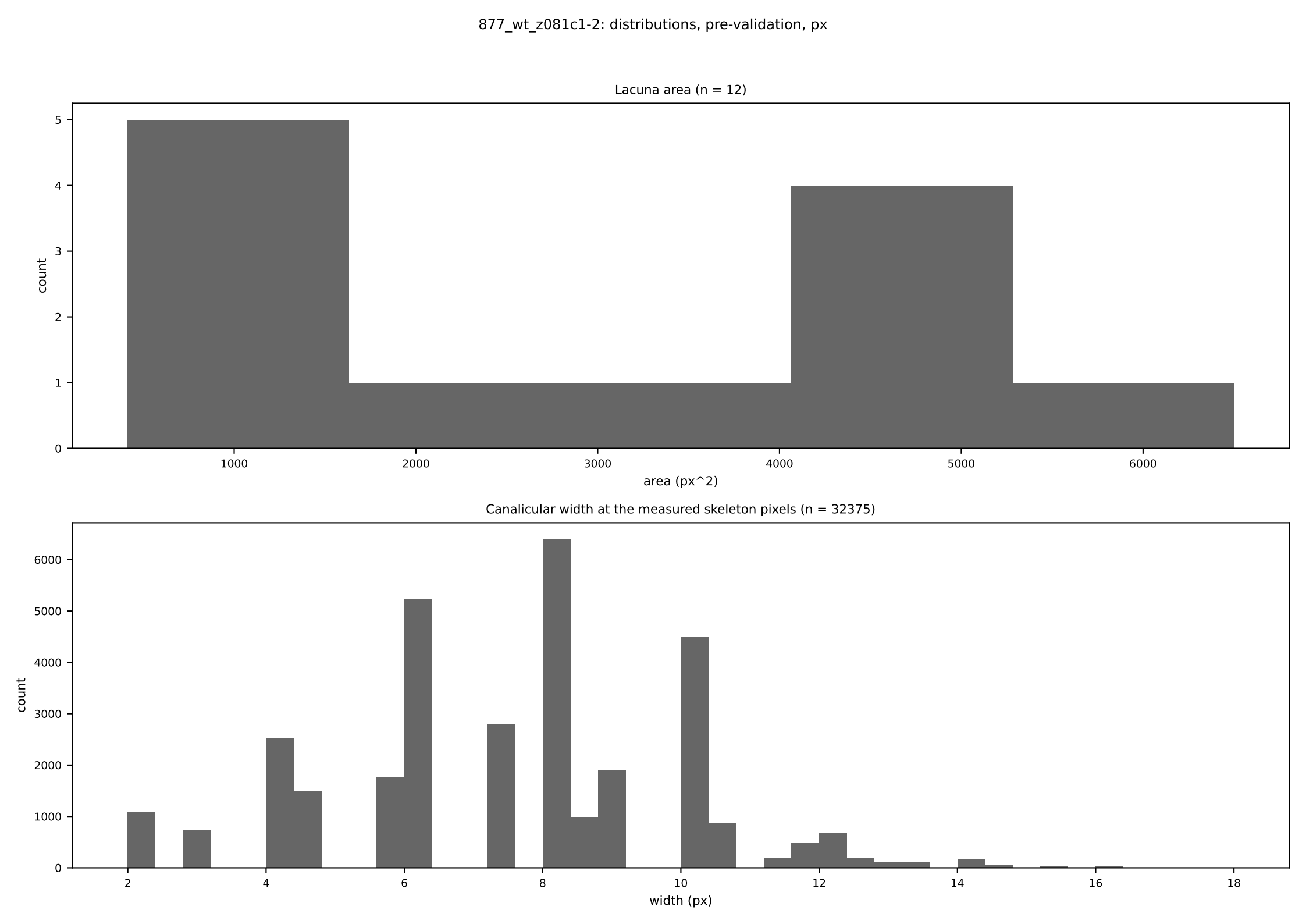
8	6	251	766	2108.3	103	20.4689	6.9792
9	17	531	1301	3493.66	175	19.9638	7.1902
10	0	65	236	0	0	0	7.9591
11	9	366	980	2777.99	128	21.703	6.7753
12	1	54	194	13.4853	1	13.4853	5.006

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	149.397	4144	11016	0.5976	0.5673	0.0542954	0.0567357	5.3549
2	351.404	9408	21117	0.7903	0.7218	0.0323129	0.0373159	2.2766
3	515.671	15609	34048	1	1	0.0461272	0.0464345	3.8784
4	108.397	5713	15341	1	1	0.0553124	0.0531908	6.4577
5	326.735	7997	18643	0.6576	0.6224	0.0308866	0.0381376	1.8364
6	354.617	12438	30402	1	1	0.0346519	0.0386817	3.1019
7	216.108	5237	12563	1	1	0.0315066	0.0432222	1.3882
8	263.451	6630	15255	1	1	0.0378582	0.050213	2.2775
9	355.262	12275	29427	1	1	0.0432587	0.0442111	4.7852
10	148.953	4761	12108	0.6788	0.6158	0.0136526	0.0194913	0
11	323.99	11381	28242	1	1	0.0321589	0.0347001	2.7779
12	85.9828	5167	15978	1	1	0.0104509	0.0121417	1.163

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	225	571	264.693	727.34	6	7	7
2	276	705	348.563	910.021	9	8	10
3	710	1495	833.767	1830.83	19	17	25
4	313	780	356.492	933.08	5	9	12
5	205	584	278.966	816.896	6	9	10
6	356	926	487.517	1347.01	10	15	16
7	108	314	190.409	624.399	3	6	9
8	219	587	282.338	873.512	3	8	12
9	466	944	611.5	1489.44	12	15	17
10	0	0	74.9411	270.451	0	4	4
11	277	592	416.019	1150.94	9	15	17
12	51	51	59.1127	215.309	2	1	1


877_wt_z081c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 32375)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16
18